

Prediction with (shallow) Neural Networks

Adapted by EVL, FRC and ASP

2025-03-17

An example using R

A predictive ANN

We use the `neuralnet` package to build a simple neural network to predict if a type of stock pays dividends or not.

```
if (!require(neuralnet))  
  install.packages("neuralnet", dep=TRUE)
```

Data for the example

And use the `dividendinfo.csv` dataset from <https://github.com/MGCodesandStats/datasets>

```
mydata <- read.csv("https://raw.githubusercontent.com/MGCodesandStats/datasets/master/dividendinfo.csv")  
str(mydata)
```

```
'data.frame':  200 obs. of  6 variables:  
 $ dividend      : int  0 1 1 0 1 1 1 0 1 1 ...  
 $ fcfps         : num  2.75 4.96 2.78 0.43 2.94 3.9 1.09 2.32 2.5 4.46 ...  
 $ earnings_growth: num  -19.25 0.83 1.09 12.97 2.44 ...  
 $ de           : num  1.11 1.09 0.19 1.7 1.83 0.46 2.32 3.34 3.15 3.33 ...  
 $ mcap         : int  545 630 562 388 684 621 656 351 658 330 ...  
 $ current_ratio : num  0.924 1.469 1.976 1.942 2.487 ...
```

Data pre-processing

```
normalize <- function(x) {  
  return ((x - min(x)) / (max(x) - min(x)))  
}  
normData <- as.data.frame(lapply(mydata, normalize))
```

Test and training sets

Finally we break our data in a test and a training set:

```
perc2Train <- 2/3  
ssize <- nrow(normData)  
set.seed(12345)  
data_rows <- floor(perc2Train * ssize)  
train_indices <- sample(c(1:ssize), data_rows)  
trainset <- normData[train_indices,]  
testset <- normData[-train_indices,]
```

Training a neural network

We train a simple NN with two hidden layers, with 4 and 2 neurons respectively.

First we define the parameters and hyperparameters

```
hidden_layers <- c(4,2) # Two hidden layers with 4 and 2 neurons  
activation_function <- "logistic" # Sigmoid  
optimizer <- "rprop+" # Optimizer  
learning_rate <- 0.01 # Learning rate  
threshold_value <- 0.01 # Convergence threshold  
max_iterations <- 1e6 # Max iterations
```

```
# Neural Network  
library(neuralnet)  
nn <- neuralnet(#dividend ~ .,  
               dividend ~ fcfps + earnings_growth + de + mcap + current_ratio,  
               data=trainset,
```